Md Muntaqim Meherab

Computer Vision & Remote Sensing Researcher | B.Sc. CSE Student

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Website GitHub LinkedIn Google Scholar

Research Profile

My research focuses on robust and computationally efficient computer vision for remote sensing and climate applications. Recent work examines mid-layer representations in frozen vision transformers, predictive confidence under cloud occlusion in flood segmentation, and robustness to noisy or missing inputs. My broader aim is to improve the reliability of vision models under limited compute and imperfect observations. I emphasize controlled evaluation and reproducible implementations.

Education

B.Sc. in Computer Science & Engineering
Daffodil International University, Dhaka, Bangladesh
Expected graduation: 30 June 2027.

Publications & Research Outputs

Main-conference and conference-proceedings papers

- **Zero-Gradient Mid-layer Evolutionary Homophilic Exploration for Remote-sensing Adaptation with Frozen Backbones** 2027
  Md Muntaqim Meherab et al.
  IEEE/CVF Winter Conference on Applications of Computer Vision. Accepted; proceedings forthcoming. Link

- **Dynamic Region-Aware Gradient Suppression (DRAGS): Enhancing Vision Model Robustness by Suppressing Noisy Feature Regions During Training** 2026
  Md Muntaqim Meherab et al.
  Proceedings of the International Conference on Intelligent Data Analysis and Applications (2025), Atlantis Press, pp. 679–687. Link

- **Multi Domain Patch-Dropout: A Simple Input-Space Regularizer That Improves Robustness Under Missing Data** 2025
  Md Muntaqim Meherab et al.
  5th International Conference on Trends in Electronics and Health Informatics (2025). Link

Workshop and bridge-program papers

- **Does Confidence Track Visibility? Diagnosing Blind Confidence in Cloud-Occluded Flood Segmentation** 2026
  Md Muntaqim Meherab et al.
  NeurIPS 2026 Workshop on Tackling Climate Change with Machine Learning. Link

- **What Your CXR Reveals Beyond the Pathology: Causal Evidence for Demographic Leakage in Medical Foundation Model Embeddings and Its Calibration Fairness Consequences** 2026
  Md Muntaqim Meherab et al.
  ECCV 2026 Workshop on Medical Foundation Models and Benchmarks. Link

- **Advancing NLP Equity: A Secondary Benchmark Evaluation of Multilingual Language Models for Underrepresented Languages** 2026
  Md Muntaqim Meherab et al.
  AAAI 2026 Workshop on Language Models for Underserved Communities. Springer CCIS version to appear. Link

- **CoSSA: Correlation-Structure Shift Adapter for Cross-City Urban Forecasting** 2025
  Md Muntaqim Meherab
  NeurIPS 2025 UrbanAI Workshop. Link

Books

- **Python Made Simple: For Everyone**. Link
  Amazon KDP, 2026.
- **Safe & Smart AI Prompts: Schema-First Prompt Engineering for LLMs**. Link
  Amazon KDP, 2025.

Awards & Achievements

- **Library Excellence Award**, Central Library, DIU: Best Library User, first position.
- **Champion Team**, Data Structures course project; awarded by the Dean, FSIT, DIU.
- **Champion Team**, Object-Oriented Programming course project: *Smart Agro System*.

Academic Service

- **Conference Reviewer:** COLM.
- **Workshop Reviewer:** Workshops at NeurIPS, ICLR, and ICML.

Leadership & Volunteer Service

- Assistant General Secretary, Daffodil AI Club.
- Executive Member, DIU Data Science Club.
- Former General Secretary, DIU Reader's Club.
- Volunteer, 4th National Data Science Summit.

Technical Skills

Programming	Python, C/C++, Java
Libraries	PyTorch, NumPy, Pandas, Scikit-learn
Computer Vision	Vision transformers, representation learning, remote-sensing image analysis, image classification, semantic segmentation
Model Evaluation	Robustness testing, confidence calibration, controlled experiments, reproducible evaluation
NLP & LLMs	Natural language processing, LLM prompting and evaluation
Tools	Git, Linux, Google Colab, Jupyter Notebook, LaTeX

References

- **Sheak Rashed Haider Noori**
  Professor & Head, Department of Computer Science & Engineering, Daffodil International University.
  Daffodil Smart City (DSC), Birulia, Savar, Dhaka–1216, Bangladesh.
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- **Nuruzzaman Faruqui**
  PhD Researcher, Monash University, Australia.
  Assistant Professor (on study leave), Department of Software Engineering,
  Daffodil International University, Bangladesh.
  Tel: +8801763341153 Email: `faruqui.swe@diu.edu.bd`
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